

Nolan Black, Ph.D. | Simulation Engineer

Redacted – U.S.A.

📧 Redacted • 📧 Redacted • 🌐 NolanBlack.github.io

Professional & Research Experience

Pasteur Labs | Scientific ML Startup

(Remote) Philadelphia, PA

Simulation R&D Engineer

January 2025 – Present

- Developed a suite of products for automated simulation, physics informed surrogates, and design optimization across CFD, structural, thermal, and multiphysics domains for aerospace, automotive, and defense industries
- Supported customer deployments and collaborations in which production engineering workflows were adapted for automation, data analysis, and surrogate development
- Designed differentiable software to perform structural optimization of pre-trained ML surrogates in aerospace applications (Python/PyTorch)
- Developed and executed simulation workflows (Ansys suite, Siemens StarCCM+, and OpenFOAM) to generate high-fidelity datasets supporting surrogate modeling for engineering design

Drexel University | GAANN Fellowship

Philadelphia, PA

Graduate Research Assistant

September 2020 – January 2025

- Developed and validated high-performance finite-element models for multiscale design optimization (C++/PETSc)
- Investigated novel methods of Machine Learning (ML) for physics-informed simulation (C++/Python/PyTorch)
- Integrated ML techniques to design nature-inspired metamaterials with tailorable physical properties

Lawrence-Livermore Nat. Lab. | Center for Design Optimization

Livermore, CA

NSF PhD INTERN

January 2024 – May 2024

- Executed high-performance computing workflows in a U.S. national laboratory environment
- Implemented parallel solvers for mixed finite element methods in a high-performance computing environment (C++)
- Developed nonlinear finite element programs for hyperelastic, hierarchical materials for applications in structural optimization

Zimmer Biomet | Zimmer Knee Creations R&D

Exton, PA

Development Engineering Coop

April 2019 – April 2020

- Developed functional medical devices using rapid prototyping to perform orthopedic procedures on the hand and wrist
- Coordinated with orthopedic surgeons to established design criteria and verify medical device designs in cadaveric lab testing
- Adapted, tested, and re-validated existing products to meet new standards of quality based on risk management

NAVSEA | Department of the Navy

Philadelphia, PA

Entry-level Engineer

April 2017 – September 2018

- Supported US Navy shipboard engineering systems through development and revision of procedures and system documentation
- Coordinated with shipboard engineering personnel to translate operational requirements into clear, auditable engineering documentation

Skills & Interests

Computational

- C++ (MPI, PETSc, mfem)
- Python (PyTorch, JAX, SciPy, NumPy)
- MATLAB

CAE Tools

- CFD: Ansys Fluent, OpenFOAM, StarCCM+
- Solid Mechanics: mfem, Ansys Mechanical
- CAD: Solidworks, Gmsh, CREO, Paraview

Education

Ph.D., Mechanical Engineering / Computer Science

Drexel University, Philadelphia, PA

September 2020 – December 2024

Thesis: Multiscale Structural Optimization through Second-order Homogenization and Machine Learning

B.S./M.S., Mechanical Engineering

Drexel University, Philadelphia, PA

September 2016 – June 2020